**Math 4400, Fall 2016** **Quiz 11** Name: ______________________________

*Calculators, cell phones, or notes are not allowed.* You may use the formula

$$(x^2+y^2)(u^2+v^2) = (xu+yv)^2 + (xv-yu)^2.$$

(1) Use the fact that 2 is a generator of $\mathbb{Z}/29\mathbb{Z}$ to find a square root of $-1$ in $\mathbb{Z}/29\mathbb{Z}$.
(2) Use (1) to find $x, m$ such that $x^2+1 = m \cdot 29$.
(3) Use the method of descent to represent 29 as a sum of two squares. (No credit given for guessing that $29 = 2^2+5^2$.)

$2^{28} = 1$, so $2^14 = -1$, so $(2^7)^2 = -1$. $2^7 = 2^5 2^2 = 32 \cdot 4 \cong_{29} 3 \cdot 4 = 12$.
Thus $12^2 + 1^2 = 145 = 5 \cdot 29$.
We have $x = 12$ and $y = 1$ and $m = 5$ so $u = 2$ and $v = 1$ and so $x_1 = (12+3)/5 = 3$ and $y_1 = (36-1)/5 = 7$. But then

$$3^2 + 7^2 = 58 = 2 \cdot 29$$

and so $u_1 = 1$ and $v_1 = 1$. But then $x_2 = (3+7)/2 = 5$ and $x_2 = (7-3)/2 = 2$. This gives $5^2 + 2^2 = 29$.